

Travel Assistant Accuracy Plan

Sep 30, 2026 · @Sayantika

Decision: ship A and B as one rule, in a 4 engineer-week V1. No price, availability or policy claim reaches the user without evidence from its source. V1 is tuned in a sandbox, V2 is chosen from what V1 measures, and Option C is cut.

The plan on one page

The assistant states facts it does not own, in one confident voice, and nothing checks a claim against its source before the user sees it. V1 adds that check.

Diagnosis	The 50% is a trip-level rate, so component errors multiply. A sizing model reproduces 50% with flight prices right only 60% of the time.
Root cause	Three failures filed as one: stale or wrongly keyed prices, policy answered from model memory, and fare attributes answered from the wrong source.
Decision	One rule, two evidence providers. A supplies live evidence for prices at the moment the user decides. B refuses policy claims without a retrieved source. C is cut.
Why not C	It moves staleness from a cache, where it has an age, into model weights, where it has none. Confidence 30%, and it uses the whole budget.
V1 capacity	4 engineer-weeks, read as 2 engineers × 2 weeks = 20 engineer-days. V1 uses 18.5; 1.5 days are held for whichever root cause week-1 data confirms.
V1 scope	Price cards bound to an offer ID, total price with fees, re-price at decision, checkout re-validation, supplier fallback, policy router rule, cite-or-deflect gate, human handoff, claim telemetry.

Expected effect	Sizing, not forecast: trip booking errors from about 50% to 25–32% after V1, depending on the cause mix, and about 11% after V2.
Left open in V1	Hotel live availability, fare jumps between decision and payment, answering meal and baggage questions (V1 deflects them), an authoritative visa feed.
How we will know	One north star (trip booking error rate, measured against the price the user acted on), three drivers, two guardrails. Tuning is config, not code.

The line for the jury: volatile facts are fetched, regulatory facts are cited, nothing is memorised, and every claim carries its evidence.

1. Context and problem breakdown

The brief lists four symptoms and one metric. Underneath sit three different failures in three different layers, and the 50% itself mixes a real error rate with a measurement effect.

1.1 What the brief says, and what each line implies

Line in the brief	What it implies
"AI assistant for travel booking... Place A to Place B, hotels or trips"	Transactional and multi-part. One trip bundles several claims that must all be right.
"Hallucination complaints have spiked in certain areas"	"Spiked" means something changed recently. "Certain areas" means the failure is not uniform. Both point to a findable cause.
"Flight prices that are wrong by the time the user clicks through"	The error is found at checkout. It is made earlier.
"Recommending hotels that are actually sold out"	Inventory freshness. A different data source from flight price.
"Confidently giving wrong visa/baggage policy info"	"Confidently" is the real defect: nothing separates a grounded answer from a guess.

Line in the brief	What it implies
"Suggesting wrong meals in the flight"	A fare- and route-specific attribute. Neither a price nor a regulation.
"50% error in the overall trip bookings, primarily due to wrong prices in flight tickets"	A trip-level metric. It compounds across components and is measured where price errors surface fastest.
"4 engineer workweeks", and under C, "may not be able to complete in 2 weeks"	Read together: about 2 engineers for 2 calendar weeks, or 20 engineer-days.
Option A con: supplier APIs "rate-limit or time out, which needs a fallback"	A cannot ship without its fallback. The fallback is part of A's scope.
Option B pro: "the highest-liability one"	The brief itself ranks by severity, not only by volume.

1.2 What we are actually solving

The assistant acts as the source of truth for facts it does not own. It sounds equally sure reading live data, a stale cache or its own memory.

The job: cut trip booking errors fast, close the stranding risk on visa answers, and not slow the assistant enough to lose bookings, all in 20 engineer-days.

Claim the assistant makes	Who owns the truth	How fast it changes	Cost if wrong
Flight price	Airline inventory, via supplier API	Minutes; offers expire in 15–30 min	Failed or abandoned booking
Hotel availability and rate	Hotel inventory, via channel manager	Minutes	Failed booking, or a guest turned away
Visa and entry rules	Destination government	Weeks; sometimes days	Denied boarding, stranding, liability
Baggage allowance	Fare family of this ticket	Per fare	Fee at the bag counter
Refund and cancellation terms	Fare rules plus our fee schedule	Per fare	Money disputes; regulatory exposure in India

Claim the assistant makes	Who owns the truth	How fast it changes	Cost if wrong
In-flight meal	Airline, per flight and cabin	Per flight	Passenger has no suitable meal
Travel concepts (what a layover is)	General knowledge	Years	Low

1.3 How the assistant works today

A request passes through intent parsing, a router that picks a data path, a cache or live supplier call or document store, prompt assembly, the model, and checkout. No step checks the reply against its source before the user sees it. The [Travel Assistant Failure Map](#) plots 40 failure modes (F-01 to F-40) onto these nine layers; this plan cites them by ID.

1.4 Why 50%: component errors multiply

A trip succeeds only if every component is right, so the rates multiply rather than average.

Component	Right today (illustrative)	Right if flight price is fixed
Flight price	60%	98%
Hotel availability	92%	92%
Fees and ancillaries	92%	92%
Whole trip	50.8%	82.9%

The inputs are illustrative, chosen to reproduce the brief's figure. What holds for any plausible inputs: the weakest component dominates, so fixing it lifts the trip rate far more than its own share suggests.

1.5 Seven mechanisms behind "wrong flight price"

The brief names price as the driver but not the mechanism. Seven mechanisms produce the same complaint, each leaves a different fingerprint in a week of logs, and each needs a different fix.

Mechanism	What happens	Fingerprint in the data	Fixed by
Quote aged before the decision	A cached or earlier fare outlives its offer	The gap grows with minutes between quote and tap	A2 re-price at decision
Fee-shaped gap	The quote excludes taxes, our convenience fee or ancillaries	The gap equals a fee, even on fresh quotes	A6, then A7
Fare bucket closed	The cheap booking class sells out and the fare steps up	Step-shaped gaps on busy dates and routes	A10 holds (V2-V3)
Wrong cache key	A fare cached for 1 adult is served to 2, or in the wrong currency	The gap tracks party size or currency, not age	A2 with real passengers; A8
Model-restated number	The model repeats an old price from the chat, rounds it, or adds a total	A price in the reply with no offer ID	A1 price cards and validator
Partial supplier results	"Cheapest" claimed while two suppliers timed out	Superlatives during timeouts	A5 fallback rules
Demand spike	Peak demand closes buckets faster and exhausts rate limits	The error spike tracks traffic and fallback rate	Measure first; supplier rate-limit plan

If the spike coincides with a demand peak, such as India's October–November festive season, much of it may be bucket depletion. That calls for holds and a faster path to payment, not a fresher cache.

1.6 Causes the brief does not name

- **Retrieval never fires.** A visa question classified as chat triggers no lookup, so the model answers from memory and no error is raised (F-04). This is the likeliest origin of the visa complaints, and Option B alone does not repair it.
- **Constraints dropped at parse.** "Jain meal for my mother" never becomes a filter, then gets answered from memory later (F-02).

- **Attribute questions sent to the wrong source.** Meals and fare-specific baggage live in supplier fare data, not in policy documents (F-05, F-25).
- **True but incomplete answers.** "No visa needed" can be correct and still strand someone who lacked an e-authorisation, an arrival card or passport validity (F-26).
- **Measurement asymmetry.** The 50% is measured at checkout, where price errors surface in minutes. Visa, meal and refund errors surface days or weeks later and are rarely traced back, so the metric over-counts price and under-counts the costliest errors.

1.7 Questions for week 0, and the assumptions used meanwhile

1. What changed 4–6 weeks ago: model or prompt version, cache TTL, a new supplier, traffic past rate limits, a new route, or seasonal demand?
2. How is a "trip booking error" defined: any price difference or above a threshold, against the first quote or the last?
3. Where do quoted prices come from today (cache, live call or model text), and in what share?
4. Is there a policy document store, and how fresh is it?
5. Does the assistant create orders itself, or hand off to a checkout page where the user confirms?

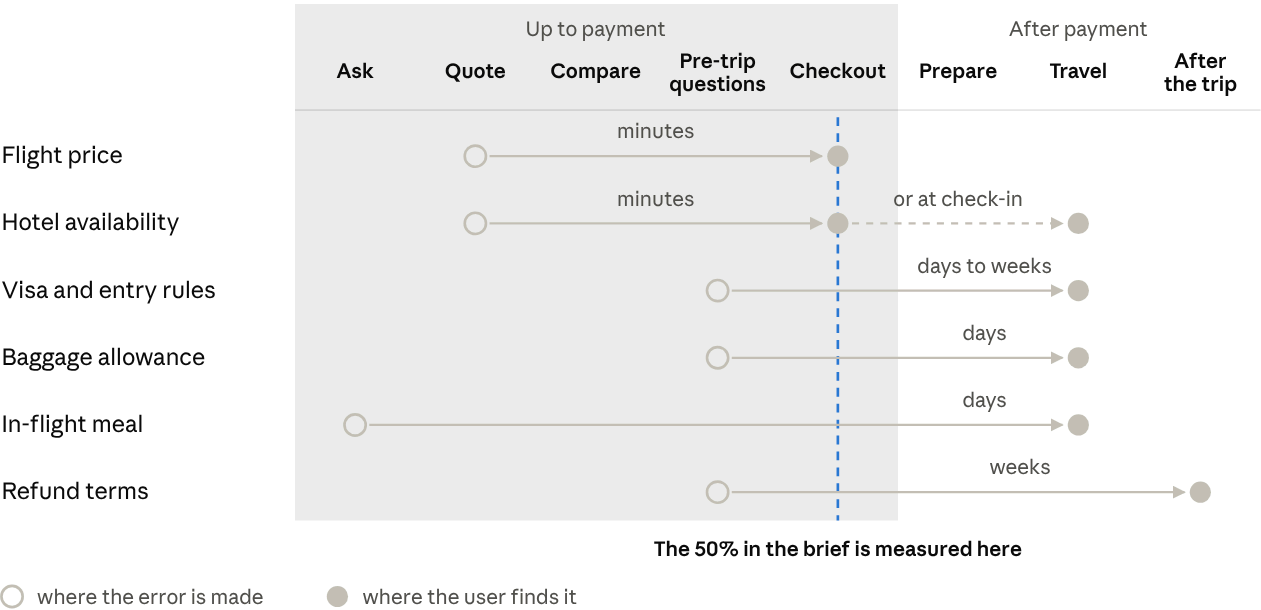
Assumption	Value used here	Why it matters
Capacity	2 engineers × 2 weeks = 20 engineer-days	Sets V1 scope
Volume	100,000 trip sessions a month reach a price quote; 40,000 reach payment	RICE reach
Mix	90% include a flight, 45% a hotel, 30% ask a policy or fare question, 12% ask about entry rules	RICE reach
Users	Mostly in India, mostly on mobile	Latency tolerance, fee rules, regulator
Checkout	The assistant hands off to a checkout page where the user confirms	Lowers write-action risk in V1

2a. The user journey, happy and unhappy

A trip passes through eight stages where the assistant makes or relies on a claim. Errors are made in the first four and mostly found in the last three, which is why complaints arrive late and get filed against the wrong step.

Errors are made early and found late

The 50% is measured at checkout, so it sees two of the six claim types



where each claim type fails · 6 claims across 8 stages

Only flight price and hotel availability fail before payment, where the 50% is counted; the other four surface days or weeks later.

Example request used below: "Delhi to Bangkok, 2 adults, 24–28 October, under ₹60,000 total, and a Jain meal for my mother."

Stage	Happy flow	Unhappy flow: the friction checkpoint	Truth owner	Found when	Map IDs
1. Ask	Every slot and constraint captured and read back: 2 adults, total budget, Jain meal	Budget read per person; party defaults to 1 adult; the Jain meal never becomes a filter	The user	Checkout (party, budget); on board (meal)	F-01, F-02, F-03

Stage	Happy flow	Unhappy flow: the friction checkpoint	Truth owner	Found when	Map IDs
2. Quote	Live offers for 2 adults with taxes and fees in the total; hotel rooms with live rate and availability	A cached 1-adult fare; base fare without fees; "cheapest" from a partial supplier set; a sold-out room shown	Supplier and hotel inventory	Checkout	F-12, F-14, F-15, F-18, F-20
3. Compare (user may leave and return)	Price re-checked when the user shortlists or comes back	A 40-minute-old price restated from the chat; the offer has expired	Supplier	Checkout	F-09, F-31, F-33
4. Pre-trip questions	Each answer quotes its source: entry rule with arrival card and stay length, bag allowance of this fare, meal for this flight and cabin, refund per fare rules	"No visa needed" with no arrival card or stay limit; a carrier-wide bag rule; "Jain meal available" on a domestic low-cost flight that sells snacks only; a generic refund rule	Government; fare family; airline catering; fare rules	Airport, on board, or weeks later	F-04, F-05, F-21, F-25, F-26, F-27
5. Checkout and confirm	Re-priced before the payment page; same price, or the change shown before paying; the user confirms the exact itinerary	The price jumps on the payment page; the offer is gone after the payment attempt; ticketing fails after payment	Supplier, transactionally	Here	F-10, F-11, F-38, F-39
6. Prepare (days before)	Visa or e-authorisation applied for in time; arrival card filed; meal request sent 24-48 h before departure; alert if a rule changed since booking	Nobody files the meal request; the entry rule changes after booking with no alert	Airline; government	Airport or on board	None on the map

Stage	Happy flow	Unhappy flow: the friction checkpoint	Truth owner	Found when	Map IDs
7. Travel: airport, flight, stay	Documents pass the airline's check; bags within allowance; meal served; hotel honours the booking; disruption help reaches a real number	Denied boarding for a missing authorisation; excess-bag fee; no Jain meal; guest turned away from an oversold hotel; a scam support number	Airline (checks against Timatic); hotel	Here, at the highest cost	F-18
8. After the trip	Refund per fare rules with the fee disclosed at booking; lost-bag claim filed at the airport with the right steps	A cancellation fee far above the airline's; an invented refund policy; compensation promised that the airline does not owe	Fare rules plus our fee schedule	Weeks later	F-14

The brief's 50% is measured at stage 5. Four of the six claim types fail after that, so the metric cannot see them and the complaint queue sees them late, if at all. V1 therefore steers those claims by leading indicators (grounded claim rate and deflection rate), because their outcome data arrives too late to steer by.

2b. What happened to others

Every documented failure below came from a model stating a fact it did not own. Every fix that held moved that fact back to a source that could be checked.

Failures in travel

Case	What happened	Matches our complaint	Lesson, and the feature it gives us
Air Canada website chatbot , ruling Feb 2024	Told a bereaved customer he could claim a bereavement fare within 90 days, after travel. The policy excluded completed travel. The tribunal held the airline liable: about C\$650 plus interest and fees.	Refund policy	The bot's answer is the company's answer, and a link beside a free-form answer did not help. → B2: answer only from the snippet.
ChatGPT, travellers to Puerto Rico , Aug 2025	Told a Spanish couple no visa was needed. They needed an ESTA and were denied boarding.	Visa	"No visa" and "no entry requirement" are different claims. → B6 entry checklist, B5 authoritative feed.
ChatGPT, Mount Misen, Japan , 2025	Gave a wrong ropeway closing time; two hikers were stranded at the summit after sunset.	Dynamic fact from memory	Opening hours are live data. → B2: no dynamic fact without a live source.
"Sacred Canyon of Humantay", Peru , Sep-Oct 2025	AI-written plans sent tourists to a place that does not exist; about US\$160 paid for the trip.	Invented inventory	Anything recommended must exist in inventory. → A1: recommendations render from inventory records.
Google AI Overviews , Aug 2025	Surfaced a scam Royal Caribbean support number; the caller was charged US\$768, later reversed.	Contact details from the open web	Support numbers and payment links are claims too. → B7 verified contact registry.

Case	What happened	Matches our complaint	Lesson, and the feature it gives us
Microsoft MSN travel guide , Aug 2023	Listed the Ottawa Food Bank as the #3 attraction, "on an empty stomach". It had passed human review.	Unchecked generation	Human review at scale is not a control. → A1 and B2: deterministic checks.
Booking.com and Accor apps in ChatGPT , test Feb 2026	Rates the AI stated did not always match the booking card; Accor priced in euros for a US user.	Wrong price	Numbers the model writes drift from the system of record. → A1 price cards; A8 currency in the cache key.
TravelPlanner benchmark , 2024	GPT-4 completed 0.6% of multi-constraint travel plans correctly.	Dropped constraint (meals)	Models do not reliably hold many constraints at once. → T2: constraints enforced in code.
India's CCPA probe of booking platforms , May 2026	Probe into cancellation fees above what airlines charge. Agoda showed ₹4,764 where Akasa Air charged ₹299; drip pricing was cited.	Refund and fee claims	In India a wrong fee is a regulatory exposure. → A6 total price; refund terms from fare rules.
CCPA fine on SpiceJet , Jul 2026	₹1 lakh for a pre-ticked loyalty box and opt-out consent on its booking site.	Booking design	An assistant that pre-selects anything inherits dark-pattern rules. → V3 booking gate with nothing pre-selected.
Traveller trust surveys , 2026	Expedia/YouGov: 8% comfortable completing a booking through AI. Booking.com: 6% fully trust AI outputs.	Trust	Trust is scarce, so each visible error costs more than one booking. → A4 honest language, A3 disclosure.

Failures in other industries

Case	What happened	Lesson → feature
Chevrolet of Watsonville chatbot , Dec 2023	A prompt made it agree to sell a 2024 Tahoe for \$1, "a legally binding offer".	The model must never originate a price or commitment. → A1
NYC MyCity chatbot , Mar 2024	Told businesses they could take workers' tips and refuse housing vouchers. The city's response was a disclaimer.	A disclaimer does not fix a wrong regulatory answer. → B2
Klarna customer service , May 2025	The CEO said the AI push "had gone too far" on cost and quality, and brought back human agents.	High-stakes cases need a human option. → B4

What worked

Precedent	How it works	Feature it backs
KAYAK.ai , Apr 2025	Real-time provider pricing in conversational search, launched as a test lab before features graduate.	A2; sandboxed rollout
Booking.com AI Trip Planner , Jun 2023	Properties shown as a visual list with Booking.com's own pricing and deep links into the booking flow; beta for part of US Genius members.	A1; staged rollout
Expedia Rapid Price Check	Confirms a chosen rate before booking; matched, changed (with the new price) or unavailable. Links expire and must not be stored.	A3, A9
Duffel offers and order errors	Offers expire in 15–30 minutes; order creation can return <code>price_changed</code> or <code>offer_no_longer_available</code> .	A2, A3

Precedent	How it works	Feature it backs
Duffel hold orders	Some airlines let you reserve now and pay later, with a price guarantee that expires at a stated time.	A10
IATA Timatic	The document-requirements database airlines check at boarding; over 500 million travellers checked a year.	B5: answer from the source the gate uses
ATPCO Routehappy amenities	Food per flight and cabin, free or paid, for about 300 airlines.	T1 meals
Hopper Price Freeze	A \$1-40 deposit holds a fare for 1-7 days and is refunded on booking.	A10

The problem list these cases give us

ID	Problem	Seen in
P1	Price stated from a stale or wrongly keyed source	Our price complaint; Booking.com/Accor test
P2	Price excludes fees that appear later	CCPA probe; fee-shaped gaps
P3	Price changes between decision and payment without warning	Our price complaint; Duffel <code>price_changed</code>
P4	Sold-out inventory recommended	Our hotel complaint
P5	Policy answered from model memory	Air Canada; Puerto Rico; our visa complaint
P6	Policy answer true but incomplete	Puerto Rico (ESTA); Thailand's arrival card
P7	Policy source out of date	Thailand's visa-free stay for Indians cut from 60 to 30 days (reported Jul 2026)
P8	Meals and baggage not tied to the fare	Our meal complaint
P9	User constraints dropped	TravelPlanner
P10	Numbers or commitments originated by the model	Chevrolet; Humantay

ID	Problem	Seen in
P11	Hours, contact numbers and other live facts from memory	Mount Misen; Google AI Overviews
P12	No human path for high-stakes cases	Klarna
P13	No way to tell which layer failed	Our 50%, with no breakdown
P14	Rules change after booking with no alert	Thailand 60 to 30 days

The Thailand change is reported in [Outlook Traveller](#); it is exactly the kind of rule a model trained earlier would state with confidence.

2c. Feature repository

Twenty-two candidate fixes cover the 14 problems, with Option C listed for comparison. A-features decompose Option A (prices and availability), B-features decompose Option B (policy), T-features cover fare attributes and the trip, and E-features make everything measurable.

ID	Feature	Solves	How it works	Precedent
E1	Claim telemetry	P13	Logs each claim's type, evidence ID, evidence age, and the gap between decision price and payment price	Failure map, layer attribution
E2	Replay sandbox	P13	200 real conversations, labelled by the PM, replayed on every change before it ships	Barnett et al. : RAG can only be validated in operation
A1	Price cards bound to an offer ID	P1, P10	Every figure renders from an offer record. A validator removes any currency figure in the model's text that has no offer ID	Booking.com planner; the ChatGPT app mismatch

ID	Feature	Solves	How it works	Precedent
A2	Re-price at the decision point	P1	A live call when the user shortlists, taps Book, or returns after the re-price window; not on every mention	KAYAK.ai; Duffel offer retrieval
A3	Checkout re-validation and disclosure	P3, P4	Re-check before the payment page; show "price changed from ₹X to ₹Y" or "no longer available, here are two alternatives"	Expedia Price Check; Duffel error codes
A4	Honest quote language	P1, P3	Each price carries a time and a promise: "₹8,400 as of 11:40, confirmed before you pay"	None needed; copy only
A5	Supplier fallback rules	P1	On timeout, show the last known price with its age, or decline. No "cheapest" claims from a partial supplier set	Option A's own stated con
A6	Quote the total	P2	Taxes and our convenience fee inside the quoted number	CCPA drip-pricing definition
A7	Late-bound supplier fields	P2	Ancillaries and supplier fields fetched with the offer, not at payment	Duffel: offer retrieval can add baggage data
A8	Cache key fix	P1	Cache keyed on passengers, cabin, currency and point of sale	Accor pricing in euros
A9	Hotel live availability	P4	Price check on the shortlisted room before calling it bookable	Expedia Rapid Price Check

ID	Feature	Solves	How it works	Precedent
A10	Price hold	P3	Use airline holds where offered, and show when the price guarantee expires	Duffel hold orders; Hopper
B1	Policy router hard rule	P5	Visa, baggage, refund, meal and contact questions always trigger retrieval; a keyword net backs up the classifier	Air Canada; MAST : reasoning-action mismatch is 13.2% of agent failures
B2	Cite-or-deflect gate	P5, P6, P8, P11	Answer only from a retrieved snippet about the right entity, quoted and linked; otherwise hand off to the official page	Option B; Air Canada; NYC MyCity
B3	Last-verified stamp and change detection	P7	Show when each source was last checked; flag when its content changes	Common practice
B4	Human handoff	P12	Visa and "I'm at the airport" cases route to a person	Klarna
B5	Authoritative entry-rules feed	P5, P6, P7	Structured visa and document data from the class of source airlines check at boarding	IATA Timatic
B6	Entry-requirement checklist	P6	Every entry answer covers visa or e-authorisation, arrival card, stay length, passport validity and transit	Puerto Rico (ESTA); Thailand's arrival card

ID	Feature	Solves	How it works	Precedent
B7	Verified contact registry	P11	Support numbers and payment links come only from a registry	Google AI Overviews scam
T1	Fare-attribute answers	P8	Baggage from the offer's fare family; meals from flight-and-cabin amenity data	Duffel; ATPCO Routehappy
T2	Constraint carry-through	P9	Parsed constraints are read back, enforced as filters, and flagged when no option meets them	TravelPlanner
T3	Post-booking change alerts	P14	Alerts on entry-rule changes, schedule changes and meal-request deadlines	Thailand 60 to 30 days
C	Option C: fine-tuned policy model	P5	Policy knowledge trained into weights	None found; it relocates staleness

3. RICE: what ships first

Severity gates first, then RICE ranks. V1 takes every Tier 0 feature that needs no outside contract, then the highest RICE scores until the 20 engineer-days run out.

Method

Term	Unit and scale used here
Reach	Trip sessions a month exposed to the problem the feature fixes (assumed volumes in 1.7)
Impact	3 prevents stranding or money lost · 2 prevents a failed or abandoned booking · 1 prevents a visible mismatch · 0.5 clarity · 0.25 marginal

Term	Unit and scale used here
Confidence	100% documented precedent and we control the mechanism · 80% precedent exists but the cause share is unknown until E1 · 50% hypothesis · 30% the evidence points against it
Effort	Engineer-weeks; 1 = 5 engineer-days
Tier 0 (gate)	Prevents an irreversible harm (stranding, denied boarding, fraud) or a legal exposure (drip pricing)

RICE measures expected value, so it cannot see irreversibility. Many ₹500 gaps outscore one stranded traveller, and that ranking would be wrong. The gate sorts by severity; RICE sorts within it.

Scores

Rank	ID	Feature	Reach / month	Impact	Confidence	Effort (eng-weeks)	RICE	Tier 0	Wave
1	A4	Honest quote language	100,000	1	80%	0.1	800,000		V1
2	A6	Quote the total	90,000	1	80%	0.2	360,000	Yes	V1
3	A1	Price cards bound to an offer ID	90,000	2	80%	0.5	288,000		V1
4	B1	Policy router hard rule	30,000	3	80%	0.3	240,000	Yes	V1
5	A3	Checkout re-validation and disclosure	40,000	2	100%	0.4	200,000		V1
6	A2	Re-price at the decision point	60,000	2	80%	0.5	192,000		V1
7	A8	Cache key fix	90,000	1	50%	0.3	150,000		V1 buffer, if data confirms
8	B2	Cite-or-deflect gate	30,000	3	80%	0.6	120,000	Yes	V1

Rank	ID	Feature	Reach / month	Impact	Confidence	Effort (eng-weeks)	RICE	Tier 0	Wave
9	A9	Hotel live availability	45,000	2	80%	0.6	120,000		V2 first
—	Option A	As written: live call on every price mention	90,000	2	80%	1.5	96,000		Comparison
10	A7	Late-bound supplier fields	90,000	2	50%	1.0	90,000		V2 branch, if fee gaps persist
11	B3	Last-verified stamp and change detection	30,000	1	80%	0.3	80,000		V1 stamp; V2 detection
12	B6	Entry-requirement checklist	12,000	3	80%	0.4	72,000	Yes	V2, with B5
13	B4	Human handoff	3,000	3	80%	0.1	72,000	Yes	V1
14	B7	Verified contact registry	5,000	3	80%	0.2	60,000	Yes	V2
—	Option B	As written: cite-or-deflect on policy	30,000	3	60%	1.0	54,000	Yes	Comparison
15	A5	Supplier fallback rules	9,000	2	80%	0.3	48,000		V1, ships with A2
16	B5	Authoritative entry-rules feed	12,000	3	80%	1.0	28,800	Yes	V2 (contract lead time)
17	T2	Constraint carry-through	40,000	1	50%	0.8	25,000		V3
18	T1	Fare-attribute answers	15,000	1	80%	0.8	15,000		V2
19	A10	Price hold	40,000	1	50%	2.0	10,000		V3
20	T3	Post-booking change alerts	10,000	2	50%	1.5	6,667		V3

Rank	ID	Feature	Reach / month	Impact	Confidence	Effort (eng-weeks)	RICE	Tier 0	Wave
—	Option C	As written: fine-tuned policy model	30,000	2	30%	4.0	4,500		Cut

RICE = reach × impact × confidence ÷ effort. E1 and E2 are not scored: they are the preconditions that turn every confidence above into a measurement.

What the scores say

- **A4 ranks first because it costs half a day.** It changes what users expect, not what the number is, so it cannot carry V1 alone.
- **Option A splits into a better package.** As written it scores 96,000 at 1.5 engineer-weeks. A1 + A2 + A3 + A5 cost 1.7 and add offer-ID binding and disclosure before payment, while calling suppliers only at decisions, not on every mention.
- **Option B as written sits at 60% confidence.** Its gate never fires when the router skips retrieval, the likely visa failure. Adding B1 lifts B1 + B2 to 80% for 0.9 engineer-weeks.
- **Option C scores 4,500,** lowest of all: 30% confidence, the whole budget, and no refresh pipeline.
- **Two Tier 0 items wait for V2 on purpose.** B5 needs a data contract with outside lead time, and B6 needs B5's structured data to be reliable. Until then B2 deflects and B4 hands off.

Dependencies

- A2 cannot ship without A5; the brief says so.
- A3 needs A1's offer IDs to compare the decision price with the payment price.
- B2 needs B1; a gate is useless if retrieval never runs.
- B6 needs B5.
- Everything needs E1, or no result can be proven.

The feedback loop across waves

Each wave ships, runs in a sandbox, and is read before the next wave is fixed. The order of V2 and V3 is a default, not a commitment: V1's telemetry can reorder it (5.6 lists the branches). Problems move from cheap and wide-reaching to costly and narrow, while Tier

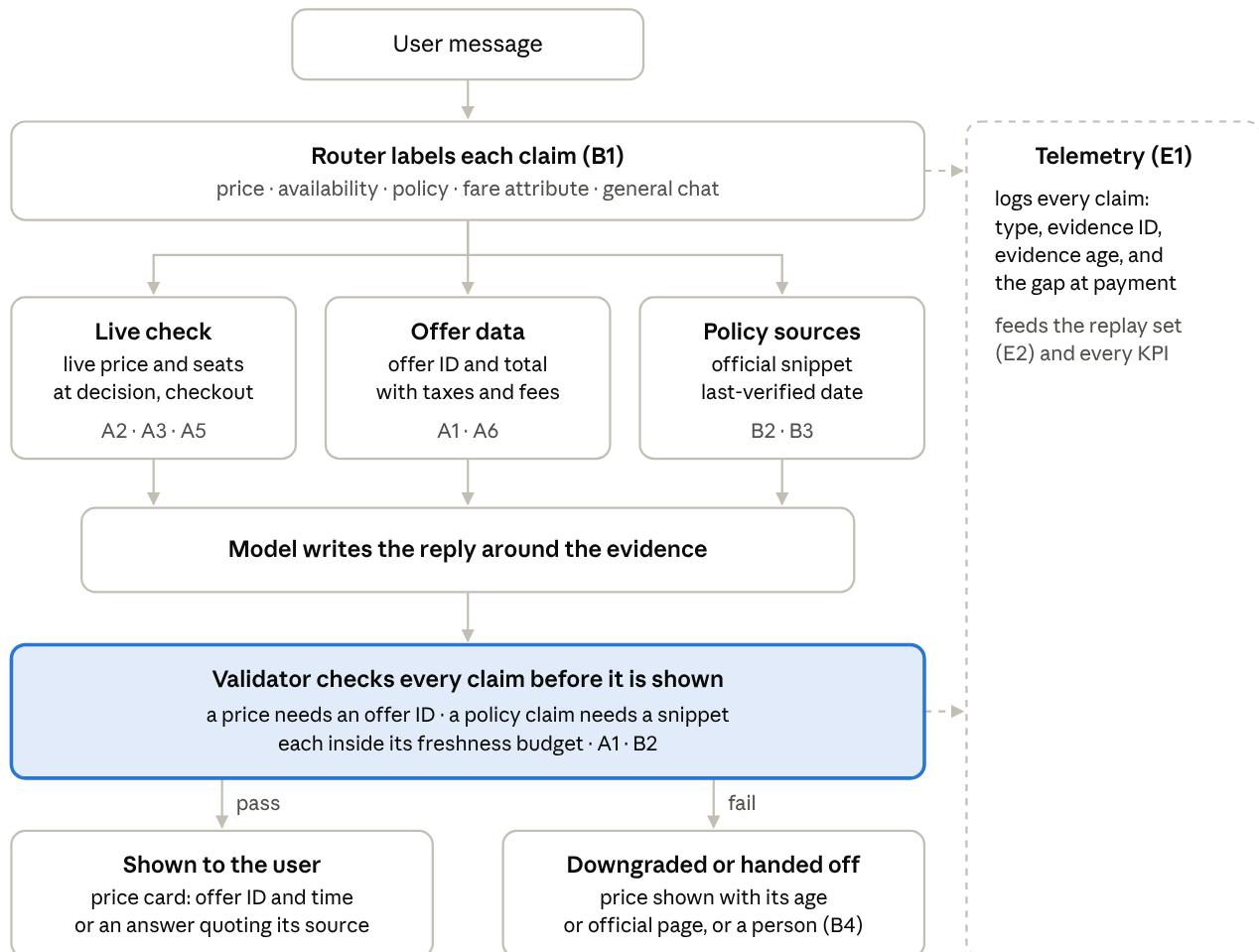
0 items jump the queue whenever they fit.

4. Architecture: what V1 closes and what it adds

V1 closes 14 of the 40 failure modes on the map and narrows 13 more; the menu as written fully covered 8. V1 also adds failure modes of its own, and one weakness sits under all of them: the claim classifier.

In V1, no claim reaches the user without its evidence

New in V1: the validator, and live checks at the moment of decision



V1 evidence path · router, three evidence sources, validator

General chat skips the evidence step, but the validator still scans every reply, so a price or policy claim cannot slip through on a mislabelled turn.

Freshness budget per claim type

Claim type	Evidence required	Budget at V1 start	If missing or too old
Flight price	Offer ID from a live supplier response	Offer unexpired, and re-checked at a decision if older than 10 minutes (K1)	Last known price with its age if under 30 minutes (K3); otherwise "let me check"
Hotel availability	Rate from the last search	Shown with its time in V1; live check arrives in V2	"Available as of 11:40, confirmed at booking"
Entry rules	Snippet from an official page for this passport and destination	Page verified within 7 days (K6)	Hand off to the official page and offer a person (B4)
Baggage, meals, refund terms	Fare data or snippet for this fare	From the current offer or its fare rules	Deflect to the fare details in V1
Contact numbers, opening hours	Registry or live source	Not available in V1	Do not state; link the official page
Travel concepts	None	None	Answer; the validator still scans for prices and policy terms

Each V1 feature: what it closes and what it can break

V1 feature	Closes	Narrows	New failure it can introduce	Health check that catches it
A1 Price cards and validator	F-31, F-33, F-39	F-34	Blocks honest ranges ("fares here usually run ₹5-6k") or misses odd formats ("5.2k", "rupees")	Block rate by pattern; weekly read of 50 blocked replies

V1 feature	Closes	Narrows	New failure it can introduce	Health check that catches it
A2 Re-price at decision	F-03, F-09, F-15, F-17	F-13, F-20	More live calls: latency, search-to-book cost, bursts on busy routes	Time to price (p90); live checks per booking
A3 Checkout re-validation		F-10, F-11, F-38	Users see more "price changed" notices, so complaints may shift before they fall	Unwarned change rate ≈ 0 ; complaints per 1,000 bookings
A5 Fallback rules	F-12, F-36 (prices)		"Last known price" becomes the silent default when a supplier degrades	Fallback rate alarm at 5%
A6 Total price		F-14	The card's fee logic drifts from checkout's fee engine	Card total vs payment total on fresh offers ≈ 0
B1 Router hard rule	F-04, F-05	F-06, F-07	False positives slow ordinary chat; false negatives still leak	Replay-set recall on policy questions $\geq 98\%$
B2 Cite-or-deflect	F-21, F-29, F-35, F-36 (policy)	F-24, F-32	A snippet exists but is the wrong one; over-deflection makes the assistant useless	Weekly audit of 50 answered policy replies; deflection band
B3 stamp only		F-27	"Verified" means fetched, not unchanged; a redesigned page silently yields no snippet	Content-hash change log; alarm on pages with no snippet
B4 handoff			Queue overflow at peak	Handoff wait time

V1 feature	Closes	Narrows	New failure it can introduce	Health check that catches it
E1 Telemetry		F-40	Logs hold nationality and passport details	Mask at write; 30-day retention

Flaws that cut across every feature

- **The claim classifier is a single point of failure.** A mislabelled claim bypasses both A and B. V1 runs two independent nets: the router decides what to fetch, and the validator scans the final text for prices, dates and policy terms whatever the router's label.
- **The metric can be gamed.** If the error rate compared the payment price with the last price shown, A3's disclosure would make every checkout match. It compares with the decision price, the one the user tapped Book on.
- **Latency stacks.** Router, retrieval, live check and validator each add time. Each step gets a budget, and the user sees "checking the live fare" instead of a blank screen.
- **Live calls cost money.** Calls fire at decisions, not mentions, and identical requests within a few seconds share one call.
- **Deflection can hide errors.** A gate that deflects everything scores 100% grounded, so deflection gets a band, not a ceiling.
- **Rollouts can contaminate results.** Users return across sessions, so the sandbox splits by user, not by session.

Still open after V1

Still open	Why V1 leaves it	Lands in
F-16 Fare bucket closes between decision and payment	Needs holds or a faster path to payment	A10 (V3), sooner if step-shaped gaps dominate
F-18, F-19 Hotel availability lag and allotments	Capacity; flights drive the 50%	A9 (V2)
F-22, F-23, F-25 Wrong or too-general snippet	Needs structured sources and audits	B5, T1 (V2)
F-26 True but incomplete entry answer	Needs structured entry data	B6 (V2)

Still open	Why V1 leaves it	Lands in
F-01, F-02 Misread slots, dropped constraints	Needs constraints enforced in code	T2 (V3)
F-28 Instructions hidden in retrieved pages	Lower frequency; hardening work	V2
F-08, F-30 Out-of-scope answers, context bloat	Lower impact on this brief	V2-V3
F-37 No gate on write actions	The assistant hands off to checkout today	V3 booking gate, before any agentic booking

5. Execution: how the team ships V1 and moves on

Two engineers build V1 in 10 working days on two independent tracks while the PM labels the replay set and writes every user-facing line. Weeks 3-4 run V1 in a sandbox where tuning is config, not code, and V2 is picked from what V1 measures.

5.1 Team and roles

Who	Owns	Days
Engineer 1, price track	A1, A6, A2, A5, A3; half of the buffer	9.5 of 10
Engineer 2, policy and platform track	E1, E2 harness, A4, B1, B2, B3 stamp, B4 handoff, flags and rollout; half of the buffer	9 of 10
PM	KPI definitions; labelling 200 replay conversations; all user-facing copy; supplier and legal conversations; daily read of shadow data; go or no-go at each rollout step	Full time

5.2 The operating loop

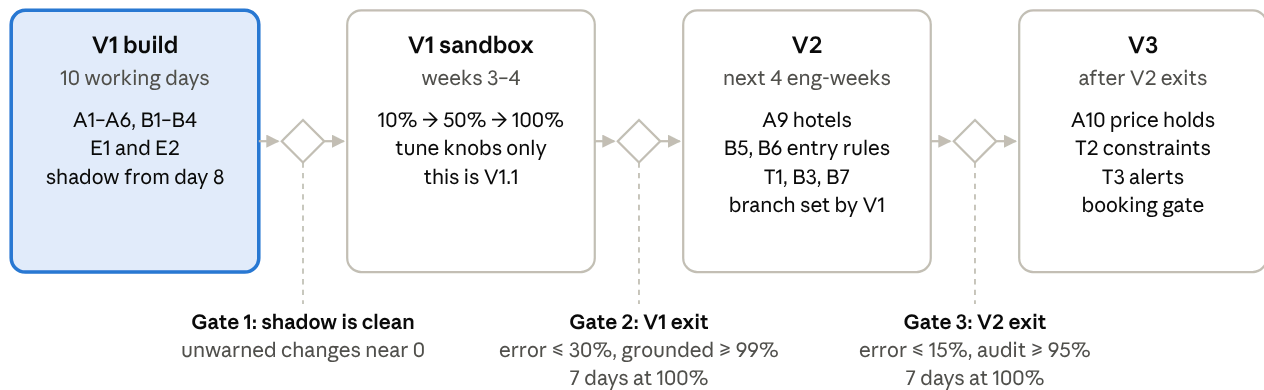
1. Build behind a feature flag.
2. Shadow: run it and log it, show users nothing, for 2-3 days.
3. Sandbox: 10% of users, then 50%, then 100%, split by user.

4. Read one north star, three drivers and two guardrails; health checks raise alarms.
5. Tune the knobs (config) and never add features mid-sandbox.
6. Graduate at the exit criteria, or roll back with the kill switch.
7. Let the same telemetry pick the next problem.

The loops overlap: while $V(n)$ is tuned, $V(n+1)$ is scoped from $V(n)$'s data, so build, tune and discovery run side by side.

Each version graduates on numbers, not on a date

$V1$ is what the 4 engineer-weeks buy; later versions wait for its numbers



A failed gate means roll back with the kill switch, tune the knobs, and run the gate again.

version roadmap · 4 phases, 3 numeric gates

Gate 2 is $V1$'s promise: it graduates only when trip errors are at or under 30% and at least 99% of claims carry their evidence.

5.3 V1 build: 10 working days

Two tracks, ten days, three decisions

Engineer 1 owns prices, engineer 2 owns policy and platform; the PM labels, writes and decides

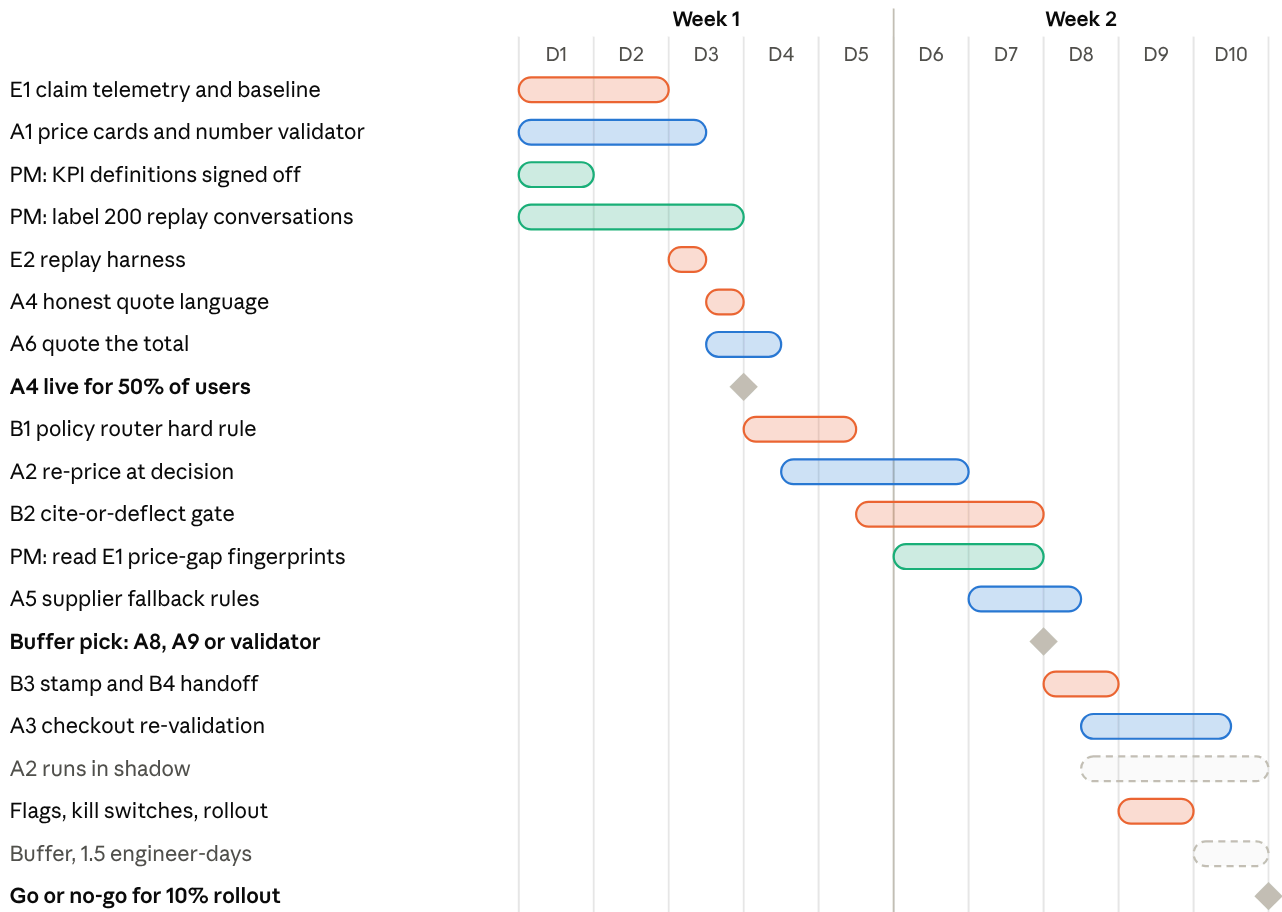
Engineer 1, price track

Engineer 2, policy and platform

PM

decision point

run or buffer



V1 build plan · 10 working days, 3 decision points

The two tracks never wait on each other: every dependency (A1 before A3, B1 before B2, A5 before A2's shadow run) stays inside one engineer's lane.

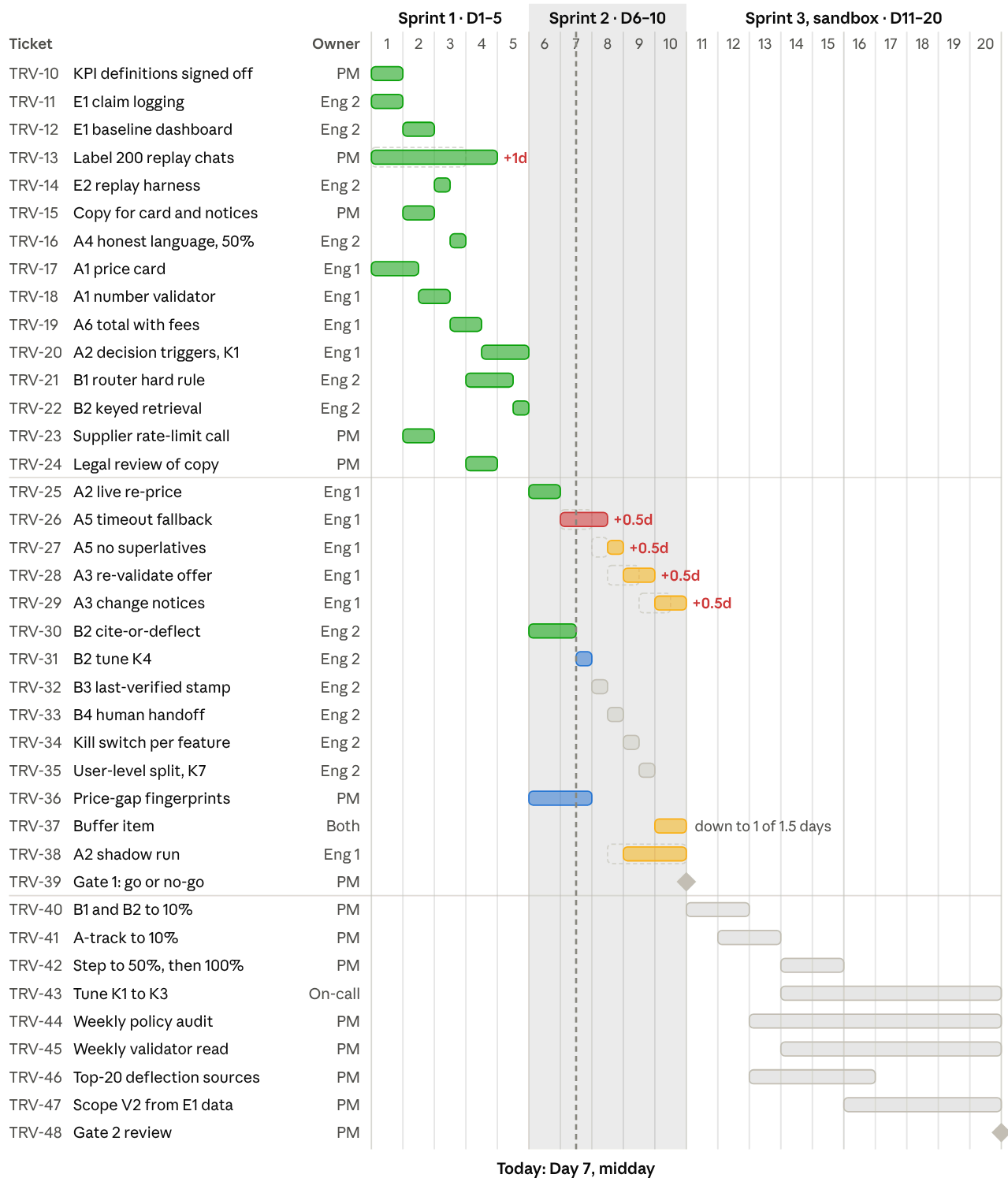
- **Ship order.** Days 1–2: E1 logs a baseline and nothing user-facing changes. Day 3: A4 to 50% of users (copy only, reversible). Days 8–10: A2 runs in shadow. Day 10: B1 and B2 go to 10% first, because they only remove risky answers; the A-track follows once shadow is clean.
- **Buffer rule, day 7.** If price gaps track party size or currency, spend the 1.5 days on A8. If hotel failures exceed 25% of trip errors, start A9. Otherwise, harden the validator.
- **Cut line.** If the build is behind on day 6, drop B3's stamp first, then the buffer item. Never drop A5 (A2 cannot ship without it) or B4 (the stranding path).

The same plan as Jira-style sprints, with every ticket, owner, point estimate and acceptance criterion: V1 sprint plan

Example Day 7: price track half a day late, buffer absorbs it

Illustrative mid-sprint status, not actuals. Bars show the forecast, dashed outlines the plan.

■ Done
 ■ In progress
 ■ Delayed
 ■ At risk
 ■ Not started
 Plan
 ◆ Gate
 +0.5d days late vs plan



V1 sprint plan tab · 39 tickets, TRV-10 to TRV-48 · example status, not actuals

In this example, supplier rate limits in staging make A5 half a day late. The four price-track tickets behind it shift into Engineer 1's half-day buffer, so Gate 1 still lands on Day 10, but the shared buffer item shrinks to one day.

5.4 V1 sandbox: knobs, KPIs and rules

V1 exposes its trade-offs as knobs, so optimising means turning a knob and reading the result, not writing code.

Knob	Start	Range	Trade-off
K1 Re-price window: re-check at a decision if the quote is older than	10 min	0–30 min	Fresher prices vs latency and live-call cost
K2 Supplier timeout	1.5 s	0.8–3 s	Complete results vs speed
K3 Fallback: show the last known price if younger than	30 min, else decline	0–60 min	Usefulness vs staleness
K4 Evidence match threshold for cite-or-deflect	Set on the replay set for $\geq 95\%$ precision	—	Fewer wrong answers vs more deflections
K5 Router sensitivity	Set for $\geq 98\%$ recall on policy questions	—	Catching every policy question vs slower chat
K6 Entry-rule freshness budget	7 days since last verified	1–14 days	Staleness vs coverage
K7 Rollout share	Shadow, then 10%, 50%, 100%	—	Risk vs speed of learning

Role	KPI	Baseline	V1 target
North star	Trip booking error rate	About 50%	$\leq 30\%$ (sizing: 25–32%)
Driver	Decision-price accuracy, flights	About 60% (implied by 1.4)	$\geq 85\%$

Role	KPI	Baseline	V1 target
Driver	Grounded claim rate	Measured in week 1	$\geq 99\%$
Driver	Policy deflection rate	0%; it never declines today	10–30% band
Guardrail	Time to price (p90)	Measured in week 1	$\leq$ baseline + 1.5 s
Guardrail	Assistant booking conversion	Measured in week 1	No drop beyond 5% relative to control

Health checks raise alarms; they are not goals. Unwarned change rate ≈ 0 . Fallback rate under 5%. Live checks per booking inside the supplier's search-to-book allowance. Validator block rate stable. Handoff wait time within the support target.

If this happens	Then do this
Decision-price accuracy rises but time to price breaches its guardrail	Raise K1 or lower K2. Add no new caching logic mid-sandbox.
Deflection goes above 30%	The corpus has gaps. The PM lists the top 20 deflected questions and adds their sources; K4 stays.
Deflection falls below 5%	The gate may be leaking. Audit 50 answered policy replies against their snippets.
Conversion drops more than 5%	Check time to price first, then deflection, and roll back the stage that moved.
The error rate stalls above 30%	Read the gap fingerprints: fee-shaped points to A7, step-shaped to A10, key-shaped to A8.
Fallback rate goes above 5%	A supplier is degrading. Share identical calls, raise it with the supplier, set K3 to decline.

Exit criteria. Seven consecutive days at 100% with the error rate $\leq 30\%$, grounded claim rate $\geq 99\%$, deflection inside 10–30%, and both guardrails holding.

5.5 V1.1: tuning only

V1.1 adds no features. Typical moves: K1 from 10 to 5 minutes on the 20 busiest routes if step-shaped gaps cluster there; sources added for the top 20 deflected questions; validator patterns fixed from the weekly sample of blocked replies.

5.6 V2: the next 4 engineer-weeks, chosen by V1's data

Default scope if the base cause mix holds: A9 hotel live availability, B5 and B6 entry-rules feed and checklist, T1 baggage and meals from fare data, B3 change detection, and B7 contact registry. That is 3.2 engineer-weeks, leaving 0.8 for the branch below.

If V1 telemetry shows	V2 leads with
Fee-shaped gaps above 30% of remaining flight errors	A7 late-bound supplier fields
Step-shaped gaps above 30%	A10 holds where airlines offer them, pulled forward from V3
Hotel failures above 25% of trip errors	A9 before anything else
Entry questions stuck above 30% deflection	B5 and B6 first
Meal and baggage questions are the most-deflected topic	T1 first

Role	KPI	V2 target
North star	Trip booking error rate	≤ 15% (sizing: about 11%)
Driver	Hotel availability failure rate	≤ 2%
Driver	Policy answer audit accuracy	≥ 95%
Driver	Policy deflection rate	5–20%, falling as sources improve
Guardrail	Time to price (p90); assistant booking conversion	As in V1

5.7 V3: holds, constraints and the trip after booking

A10 price holds, T2 constraint carry-through, and T3 post-booking alerts. Before any agentic booking, a confirmation gate: the user confirms the exact itinerary, passengers and total, and nothing is pre-selected. Option C returns only as a phrasing layer over retrieved text, never as a source. V3 adds one lagging KPI, the post-booking issue rate, to finally measure the slow-discovery claims.

5.8 What V1 knowingly does not fix

Not in V1	Why	Mitigation in V1	Lands in
Hotel live availability	Capacity; flights drive the 50%	Availability shown with its time, "confirmed at booking"	V2
Fare jumps between decision and payment	Needs holds or a faster path to payment	Disclosed before the payment page (A3)	V2-V3
Answering meal and baggage questions	Needs fare and amenity data	Deflected to the fare details (B2)	V2
Authoritative visa feed	Outside contract, with lead time	Cite-or-deflect plus human handoff	V2
Option C	Moves staleness into weights; 30% confidence; the whole budget	—	Not planned

KPI definitions

Every KPI here is a product metric: it is computed from what users see and do, never from model internals. The price tolerance (₹100 or 1%, whichever is larger) is a starting value to agree with finance.

KPI	Definition	Measured from	Used in
Decision price (term)	The total on the price card at the moment the user tapped Book	A1 card render and tap event	All price KPIs

KPI	Definition	Measured from	Used in
Trip booking error rate	Assistant checkouts where a booked component's payment price differs from its decision price beyond tolerance, or a component is unavailable ÷ assistant checkouts reaching payment	Checkout events joined to E1 claims	North star, V1-V3
Decision-price accuracy, flights	Flight checkouts whose payment price is within tolerance of the decision price ÷ flight checkouts reaching payment	Same join	V1 driver
Grounded claim rate	Price, availability, policy and attribute claims shown with valid evidence inside their freshness budget ÷ all such claims shown	Validator log	V1 driver
Policy deflection rate	Policy and fare questions answered with a hand-off to an official page ÷ policy and fare questions	Router and gate log	V1-V2 driver
Time to price (p90)	90th percentile of seconds from the user's price request to the price card rendering	Client timing	Guardrail
Assistant booking conversion	Payment confirmations ÷ trip sessions that reached a quote, treatment vs control	Funnel events	Guardrail
Hotel availability failure rate	Hotel selections that fail as unavailable at payment ÷ hotel selections reaching payment	Checkout events	V2 driver

KPI	Definition	Measured from	Used in
Policy answer audit accuracy	Answered policy replies judged correct and complete against their source ÷ replies audited (weekly sample of 50)	PM audit	V2 driver
Post-booking issue rate	Document, bag or meal problems reported per 1,000 assistant trips	Support tickets tagged by claim type	V3
Complaints per 1,000 bookings	Hallucination complaints ÷ assistant bookings × 1,000	Support tickets	Read alongside; never a target

Health check	Definition	Healthy when
Unwarned change rate	Checkouts whose payment price differs from the decision price with no notice before the payment page ÷ checkouts reaching payment	≈ 0
Fallback rate	Live checks answered by the fallback path ÷ live checks	Under 5%
Live checks per booking	Live supplier price calls ÷ bookings	Inside the supplier's search-to-book allowance
Validator block rate	Replies in which the validator removed a claim ÷ replies	Stable week on week
Handoff wait time	Minutes from handoff to a person's first reply	Within the support target

Sources

Volumes, cause shares and targets are working assumptions, labelled where used. Everything else comes from the case brief or the pages below, each opened for this plan.

- Case brief: Glance26 PM Intern case study (project file)
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- [Gulf News: ChatGPT visa advice and the Puerto Rico flight](#)
- [Great Walks: the perils of letting AI plan your trip \(Mount Misen\)](#)
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